The Complete Crouzeix Theorem

Metric Certificates, Optimal Similarity, and a Common Positive Boundary Representation

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M631

October 8, 2026

Abstract

We examine the September 23, 2026 source proof of the complete Crouzeix theorem through its finite-dimensional metric optimization and common positive boundary density. The source's universal constant two is inherited, not claimed as a contribution here. We give complete local proofs of a Stein repair with operator-model error, a trace-sensitive dual residual bound for the optimal condition number, quantitative endpoint support control, and positive finite boundary approximations with an explicit, untruncated analytic bias. Physical metrics and certified faces are treated separately: an interior repair need not preserve a singular optimal Stein slack. An exactly solvable nilpotent family verifies optimality and shows why stability does not mean strict contractivity. The read scope, private proof dependencies, source identity, and limitations of numerical checks are recorded. Contribution assessment: C3/5, a proved conditional synthesis of metric certificates and positive approximation for this source, without a new universal bound or a global novelty claim.

1 Scope, conventions, and the inherited theorem

The object of this report is the source *The complete Crouzeix theorem: optimal similarity and a common positive boundary representation* [1]. It is distinct from the direct coefficient proof examined in M630. Here the added results concern the metric semidefinite program (SDP), its certificates, and the common positive representation. They do not repeat M630's coefficient-energy defects or finite Faber Gram analysis.

All unqualified matrix norms are Euclidean operator norms. The Frobenius norm is denoted by $\|\cdot\|_F$; $X \succeq 0$ means Hermitian positive semidefinite. Tensor factors are ordered base space first and coefficient space second: if $v(z) = \sum_k z^k V_k$, then

$$v[A] = \sum_k A^k \otimes V_k, \qquad \operatorname{vec} X = \sum_j X e_j \otimes e_j, \qquad (B \otimes C)\operatorname{vec} X = \operatorname{vec}(BXC^{\mathsf{T}}).$$

The transpose here is not the adjoint. The normalized circle measure is $d\sigma(e^{i\theta}) = d\theta/(2\pi)$.

An admissible domain Ω is bounded and convex with regular analytic Jordan boundary. Fix an interior conformal map $f : \Omega \to \mathbb{D}$ and an exterior conformal map G, with $G(t) = at + b + O(t^{-1})$, $a > 0$. Both maps and their inverses extend univalently across their boundary circles. Put

$$q(t) = tG'(t), \quad \psi(t) = f(G(t)), \quad j(t) = \frac{tG'(t)f'(G(t))}{\psi(t)} > 0 \quad (t \in \mathbb{T}).$$

The last number is the derivative of the positively oriented boundary angle. If $W(A) \subset \Omega$, then $T = f(A)$ has spectral radius less than one. For any stable finite matrix T, define

$$p(T) = \min\{\tau : I \preceq H \preceq \tau I,\ \mathcal{L}_T(H) := H - T^*HT \succeq 0\}, \qquad \kappa(T) = \sqrt{p(T)}. \tag{1}$$

Theorem 1.1 (Inherited source theorem)**.** *For the admissible pair A, Ω, the minimum in* (1) *is attained and $1 \le p(f(A)) \le 4$. For an optimizer H, let $S = H^{1/2}$, $A' = SAS^{-1}$, and $D = STS^{-1} = f(A')$. Then* $\operatorname{cond} S = \kappa(T)$, *$\|D\| \le 1$, and $\rho(D) < 1$. There is one continuous density $\Lambda : \mathbb{T} \to M_n^+$, with $\int \Lambda \, d\sigma = I$, such that, for every finite m and every M_m-valued v holomorphic on a neighborhood of $\overline{\Omega}$,*

$$v[A'] = \int_{\mathbb{T}} \Lambda(t) \otimes v(G(t)) \, d\sigma(t), \qquad \|v[A]\| \le \kappa(T) \max_{\overline{\Omega}} \|v\| . \tag{2}$$

The similarity and density do not depend on m or v. For an arbitrary bounded operator A on a nonzero Hilbert space, $\overline{W(A)}$ is a complete 2-spectral set for functions holomorphic near that compact set.

The final Hilbert-space assertion does not assert the existence of a limit of the finite optimal metrics or densities. The finite representation and the infinite-dimensional inequality have different quantifiers. In particular, this report does not infer a finite positive density for an arbitrary infinite-dimensional operator.

As of the October 8, 2026 public check, the related arXiv records are 2608.03841, version 2 of August 17, and 2608.27346, version 4 of September 21 [2, 3]. These records are context, not independent verification of [1]. Classical background includes Stein stability [4], Paulsen's similarity theorem [5], and positive operator-valued representations [6]. The present source supplies its own finite separation and realization argument. We use that argument below rather than silently replacing it by a scalar-to-complete inference.

2 Metric optimization and the common density

2.1 Feasibility, attainment, and the exact optimization meaning

Proposition 2.1 (Stable-matrix metric problem)**.** *For every finite T with $\rho(T) < 1$, the SDP* (1) *is strictly feasible, has a minimizer, and satisfies*

$$\kappa(T) = \inf\{\operatorname{cond} R : R \text{ invertible},\ \left\|RTR^{-1}\right\| \le 1\}.$$

Every minimizing H has $\operatorname{cond}(H^{1/2}) = \kappa(T)$.

Proof. For $\rho(T) < \eta < 1$, Jordan form gives $\left\|T^k\right\| \le C_\eta \eta^k$. Hence

$$K = \sum_{k=0}^{\infty} (T^*)^k T^k, \qquad K \succeq I, \qquad \mathcal{L}_T(K) = I$$

converges in norm. Taking $H = cK$ with $c > 1$, and $\tau > \|cK\|$, makes all three slacks positive definite. A nonempty objective sublevel has $I \preceq H \preceq \tau_0 I$ and $1 \le \tau \le \tau_0$, so is compact. It contains a minimizer.

If R makes T a contraction, then R^*R satisfies the Stein inequality. Dividing it by its smallest eigenvalue makes (1) feasible with objective $(\operatorname{cond} R)^2$. Conversely a feasible H gives the contraction $H^{1/2}TH^{-1/2}$ and condition number at most $\sqrt{\tau}$. Applied to a minimizer, these two directions force equality. □

Strict feasibility is a property of the feasible set, not of its optimizer. An optimal H can satisfy $\operatorname{rank} \mathcal{L}_T(H) < n$, even when $\rho(T) = 0$; see Section 6.

2.2 Positive density and amplification, proved directly

Proposition 2.2 (Disk density)**.** *Suppose $\|D\| \le 1$ and $\rho(D) < 1$. For $\zeta \in \mathbb{T}$, set*

$$\Phi_D(\zeta) = (I - D/\zeta)^{-1} + (I - D/\zeta)^{-*} - I. \tag{3}$$

It is continuous and positive, has integral I, and

$$\int_{\mathbb{T}} \zeta^k \Phi_D(\zeta)\, d\sigma(\zeta) = D^k \quad (k \ge 0).$$

For every finite coefficient size and analytic matrix function h near $\overline{\mathbb{D}}$,

$$h[D] = \int_{\mathbb{T}} \Phi_D(\zeta) \otimes h(\zeta)\, d\sigma(\zeta), \qquad \|h[D]\| \le \max_{\mathbb{T}} \|h\|\,.$$

Proof. Write $E = I - D/\zeta$. Direct multiplication, in its given order, gives

$$E^*\Phi_D(\zeta)E = E^* + E - E^*E = I - D^*D \succeq 0. \tag{4}$$

Stability makes E invertible even if $\|D\| = 1$. The power estimate used above yields absolute uniform convergence of

$$\Phi_D(\zeta) = I + \sum_{k\ge1} \big(\zeta^{-k} D^k + \zeta^k (D^*)^k\big).$$

Integration gives the asserted moments, including the mass. The Taylor series for h converges absolutely on $\mathbb{T}$, so termwise integration proves the representation. For any coefficient size m, the map

$$V_m b(\zeta) = (\Phi_D(\zeta)^{1/2} \otimes I_m)b$$

is an isometry. Multiplication by $I_n \otimes h(\zeta)$ has norm at most $\max_{\mathbb{T}} \|h\|$. Its compression by V_m is $h[D]$. This proves the bound for every m using the same density. □

Applying the change of variables $\zeta = \psi(t)$ to this proof gives

$$\Lambda(t) = j(t)\Phi_D(\psi(t)). \tag{5}$$

With $h = v \circ f^{-1}$, the composition and similarity rules for holomorphic functional calculus prove (2) for any feasible H, with $\operatorname{cond} S$ in place of $\kappa(T)$. Only the universal estimate $p(T) \le 4$ remains source-specific.

Proposition 2.3 (Rank of the common density)**.** *For every $\zeta \in \mathbb{T}$,*

$$\operatorname{rank} \Phi_D(\zeta) = \operatorname{rank}(I - D^*D).$$

Consequently it is positive definite everywhere exactly when $\|D\| < 1$. If $\|D\| \le q < 1$, then

$$\frac{1-q}{1+q} I \preceq \Phi_D(\zeta) \preceq \frac{1+q}{1-q} I.$$

The corresponding ranks for Λ are unchanged; its scalar upper and lower bounds also contain the positive factor $j(t)$.

Proof. Rank and definiteness follow from the invertible congruence (4). For the lower bound, $\Phi_D \succeq (1-q^2)E^{-*}E^{-1} \succeq (1-q^2)(1+q)^{-2}I$. The Fourier series gives $\|\Phi_D\| \le 1 + 2\sum_{k\ge1} q^k = (1+q)/(1-q)$. □

3 How the source obtains the constant two

This section reconstructs the dimension-independent bridge in [1]. Its ingredients and constant are inherited. It is included to expose the hypotheses used by the certificates later.

3.1 Endpoint multipliers and a matrix extremal

For an optimizer (p, H), finite-dimensional separation with the strictly feasible point from Proposition 2.1 gives

$$X_0, Y_0, Z_0 \succeq 0, \quad \operatorname{tr} X_0 = 1, \quad X_0 - Y_0 = Z_0 - TZ_0T^*, \quad \operatorname{tr} Y_0 = p, \tag{6}$$

and the three zero products

$$X_0(pI - H) = 0, \qquad Y_0(H - I) = 0, \qquad Z_0\mathcal{L}_T(H) = 0.$$

Here is the separation detail. The affine set $(s, J - I, sI - J, J - T^*JT)$ misses the open convex set with $s < p$ and all three matrix coordinates positive definite. A separating functional has the form $\lambda s - \operatorname{tr}(Y_0U) - \operatorname{tr}(X_0V) - \operatorname{tr}(Z_0W)$, with $\lambda \ge 0$ and PSD matrix coefficients. Constancy on the affine set gives $\lambda = \operatorname{tr} X_0$ and its displayed matrix balance. If $\lambda = 0$, evaluation at a strictly feasible point contradicts nontriviality, so normalize $\lambda = 1$. Approaching the optimal point from the open set forces equality of objectives and vanishing of each nonnegative slack pairing. For PSD matrices, zero trace pairing implies zero product by taking positive square roots.

Put $S = H^{1/2}$, $D = STS^{-1}$, $p = \kappa^2$. If $\kappa > 1$, choose square factors

$$XX^* = \kappa^{-2}SX_0S, \quad YY^* = \kappa^{-2}SY_0S, \quad Q = \kappa^{-2}SZ_0S.$$

Then $\|X\|_F = \|Y\|_F = 1$, $SX = \kappa X$, $SY = Y$, and $X^*Y = 0$ as a whole matrix. Indeed the columns belong to the orthogonal endpoint eigenspaces of S. The zero Stein product is consistent with the following balance; directly, $D(SZ_0S)D^* = STZ_0T^*S$. Thus

$$XX^* - YY^* = Q - DQD^*. \tag{7}$$

Let $L = Q^{1/2}$. The rows $[L\ Y]$ and $[DL\ X]$ have equal Gram matrices. The induced isometry between the ranges of their adjoints extends to a unitary $U = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ on $\mathbb{C}^{2n}$, with

$$[L\ Y] = [DL\ X]U, \qquad L = DLa + Xc, \quad Y = DLb + Xd.$$

As $\|a\| \le 1$ and D is stable, iteration gives

$$Y = Xd + \sum_{j\ge 0} D^{j+1} X c a^j b.$$

The rational function $R(\zeta) = d + \zeta c(I - \zeta a)^{-1} b$ is contractive in $\mathbb{D}$: with $v = (I - \zeta a)^{-1} be$, $U(\zeta v, e) = (v, R(\zeta)e)$, and preservation of squared norms gives $\|R(\zeta)e\|^2 = \|e\|^2 - (1 - |\zeta|^2)\,\|v\|^2$. Every apparent boundary pole is removable, since a genuine rational pole would be unbounded along an interior radius. Hence $F_D = R^{\mathsf{T}}$ is analytic near $\overline{\mathbb{D}}$ and contractive. Transpose preserves singular values. The vectorization convention now proves

$$F_D[D] \operatorname{vec} X = \operatorname{vec} Y.$$

Writing $F = F_D \circ f$, $x = \operatorname{vec} X$, $y = \operatorname{vec} Y$, we have $F[A']x = y$. Notice that F_D is rational, but F need not be.

3.2 Exterior projection with its actual adjoint

Write the exterior expansions

$$\frac{q(t)}{G(t) - z} = \sum_{k\ge 0} b_k(z) t^{-k}, \quad b_0 = 1, \tag{8}$$

$$B(w,t) = \frac{q(t)}{G(t) - G(w)} - \frac{t}{t - w} = \sum_{i,j\ge 1} \beta_{ij} w^{-i} t^{-j}. \tag{9}$$

The coefficients b_k are polynomials: expanding at infinity in t, the coefficient at order k involves only finitely many powers of z. The diagonal singularity in B is removable. Its value tends to zero when either variable tends to infinity. Holomorphy in a product exterior collar and Cauchy estimates give, for some $r < 1$, $|\beta_{ij}| \le C r^{i+j}$. Comparison of coefficients gives $b_j(G(w)) = w^j + \sum_{i\ge 1} \beta_{ij} w^{-i}$.

The real kernel $m(w,t) = 1 + B(w,t) + \overline{B(w,t)}$ is nonnegative and both its marginals equal one. Indeed, off the diagonal it equals $2\operatorname{Re}(q(t)/(G(t) - G(w)))$; the supporting line at $G(t)$ proves positivity, which extends to the diagonal by continuity. Integrating the displayed double series proves each marginal assertion. Its integral operator $\mathcal{M}$ is an L^2 contraction by Jensen followed by the other marginal identity. This is true for matrix-valued functions with the Hilbert–Schmidt norm, without a coefficient-size factor. It sends strictly positive Fourier modes to negative ones and strictly negative modes to positive ones.

Let $u = u_0 + u_+ + u_-$ be its orthogonal Fourier decomposition. The interior Cauchy projection on boundary traces is

$$\mathcal{C}u = u_0 + u_+ + \mathcal{M}u_+, \qquad \mathcal{C}^* u = u_0 + u_+ + \mathcal{M}^* u_-.$$

The latter is the Hilbert-space adjoint, not complex conjugation of the former. In particular,

$$\|\mathcal{C}^* u\|_2^2 + \|u_0\|_F^2 \le 2\,\|u\|_2^2\,. \tag{10}$$

This follows from $\|u_+ + \mathcal{M}^* u_-\|^2 \le 2(\|u_+\|^2 + \|u_-\|^2)$. Define $\widetilde{\mathcal{C}}u = (\mathcal{C}(u^*))^*$. Its adjoint is $\widetilde{\mathcal{C}}^* u = (\mathcal{C}^*(u^*))^* = u_0 + u_- + \mathcal{M}^* u_+$.

For finite Laurent $u = \sum U_k t^k$, its interior Cauchy integral is $c_u(z) = \sum_{k\ge 0} b_k(z) U_k$, and $\mathcal{C}u = c_u \circ G$. If F is an analytic matrix trace, then

$$\mathcal{C}(F\mathcal{C}u) = F\mathcal{C}u, \qquad \mathcal{C}((\mathcal{C}u)F) = (\mathcal{C}u)F. \tag{11}$$

For Laurent input these are Cauchy's formula for the two ordered analytic products. Laurent approximation in the boundary collar justifies the formula for their analytic traces. Boundedness of $\mathcal{C}$ and of multiplication by F, followed by density of finite Laurent polynomials, extends it to all L^2 inputs. There is no assertion that every L^2 boundary function extends holomorphically across the boundary.

3.3 Positive blocks and the dimension-free comparison

The positive density of Section 2, with the extremal x, y, F, gives fields $x_\Lambda = (\Lambda^{1/2} \otimes I)x$, $y_\Lambda = (\Lambda^{1/2} \otimes I)y$. Equality in the norm-one compression $F[A']x = y$ implies

$$y_\Lambda = (I \otimes F)x_\Lambda, \qquad x_\Lambda = (I \otimes F^*)y_\Lambda$$

almost everywhere. For the first equality, equality in the isometric compression forces the multiplied field into its range. For the second, the integral of $x_\Lambda^*(I - I \otimes F^*F)x_\Lambda$ is zero; its nonnegative integrand vanishes almost everywhere.

Taking partial traces over the base space, set

$$p_0(t) = \mathrm{Tr}_{\mathrm{base}}(x_\Lambda x_\Lambda^*), \quad q_0(t) = \mathrm{Tr}_{\mathrm{base}}(y_\Lambda y_\Lambda^*), \quad r(t) = \mathrm{Tr}_{\mathrm{base}}(y_\Lambda x_\Lambda^*).$$

Here p_0, q_0 are names for fields, not their Fourier means; use $\langle u \rangle = \int u \, d\sigma$ for a mean in this subsection. Then $p_0 = F^* r$, $r = F p_0$, $q_0 = r F^*$, and

$$\langle r \rangle = (X^* Y)^{\mathsf{T}} = 0, \qquad \mathrm{tr}\langle p_0 \rangle = 1.$$

The full matrix zero is essential.

For the original A, define

$$K_A(t) = q(t)(G(t)I - A)^{-1}, \qquad E_A = K_A + K_A^*.$$

The supporting half-plane of $W(A)$ implies $E_A \succeq 0$: congruence by $G(t)I - A$ reduces this to the supporting-line inequality for its real part. Its integral is $2I$, by (8). Form the analogous fields p_e, q_e, r_e using $E_A^{1/2}$ in place of $\Lambda^{1/2}$. Their pointwise block $\Delta = \left(\begin{smallmatrix} p_e & r_e^* \\ r_e & q_e \end{smallmatrix}\right)$ is PSD, since it is a partial trace of a positive block outer product.

For finite Laurent u, coefficient integration gives exactly

$$\int E_A \otimes u \, d\sigma = c_u[A] + c_{u^*}[A]^*.$$

Similarity acts on the analytic cross pairing by κ, on the adjoint cross pairing by κ^{-1}, and on each diagonal pairing by one, because $(S \otimes I)x = \kappa x$ and $(S \otimes I)y = y$. For example $y^* c_u[A] x = \kappa y^* c_u[A'] x$.

Testing the three fields against finite Laurent u, and then using density, therefore proves the exact identities

$$\begin{aligned} p_e &= \mathcal{C}^* p_0 + \widetilde{\mathcal{C}}^* p_0, \\ q_e &= \mathcal{C}^* q_0 + \widetilde{\mathcal{C}}^* q_0, \\ r_e &= \kappa \mathcal{C}^* r + \kappa^{-1} \widetilde{\mathcal{C}}^* r. \end{aligned} \tag{12}$$

Put $g = \mathcal{C}^* r$, $P = \mathcal{C}^* p_0$, $Q_1 = \mathcal{C}^* q_0$. Taking adjoints of both ordered module identities (11) gives

$$P = \mathcal{C}^*(F^* g), \qquad Q_1 = \mathcal{C}^*(g F^*).$$

Since P, Q_1 have nonnegative frequencies and Hermitian means, (10) proves

$$\|p_e\|_2^2 = 2\big(\|P\|_2^2 + \|\langle P \rangle\|_F^2\big) \le 4\,\|g\|_2^2\,, \qquad \|q_e\|_2^2 \le 4\,\|g\|_2^2\,.$$

Also $g \neq 0$, for otherwise $P = 0$, contradicting $\operatorname{tr}\langle p_0 \rangle = 1$. Since r has zero mean, g is strictly positive-frequency and $\widetilde{\mathcal{C}}^* r$ is strictly negative-frequency. Thus $\|r_e\|_2^2 \ge \kappa^2\,\|g\|_2^2$. For $J = \operatorname{diag}(I, -I)$, positivity of Δ yields

$$0 \le \operatorname{tr}(\Delta J \Delta J) = \|p_e\|_F^2 + \|q_e\|_F^2 - 2\,\|r_e\|_F^2\,.$$

Indeed the trace on the left is the trace of the square of the Hermitian matrix $\Delta^{1/2} J \Delta^{1/2}$. Integration gives $2\kappa^2\,\|g\|_2^2 \le 8\,\|g\|_2^2$, hence $\kappa \le 2$. The case $\kappa = 1$ was already immediate. Every norm in this comparison is Hilbert–Schmidt before integration; there is no dimension-dependent conversion to a scalar entry norm.

3.4 Boundary regularization and Hilbert-space quantifiers

For completeness, the source's passage beyond admissible domains can be expressed without taking a metric limit. If $K \subset \mathbb{C}$ is a nonempty compact convex set, convolve its distance function with a two-dimensional Gaussian of scale δ and add $\delta|z|^2$. The resulting proper real-analytic function h_δ is strongly convex. If $c = \mathbb{E}|Z|$ and $R_K = \max_K |z|$, then

$$h_\delta|_K \le c\delta + \delta R_K^2, \qquad h_\delta(z) \ge d_K(z) - c\delta.$$

Choosing both errors below $\epsilon/2$ makes the sublevel $\Omega_\epsilon = \{h_\delta < \epsilon/2\}$ contain K and have closure in $K_\epsilon = \{d_K \le \epsilon\}$, after decreasing δ strictly if needed. Its level lies above the unique minimum, so its boundary is regular analytic; strict convexity and properness make it a Jordan curve. Gaussian domination justifies analytic differentiation. The analytic Jordan version of conformal reflection supplies the collars used above; the source proves it in its appendix, using local harmonic reflection and compactness. For a fixed matrix polynomial, apply the admissible result and let $\epsilon \downarrow 0$. Continuity of its maximum proves the bound on $W(A)$ for finite A. The matrix $\left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right)$, whose numerical range is the closed unit disk, proves sharpness using $v(z) = z$.

If A acts on any nonzero Hilbert space, v is a matrix polynomial of degree d, and $\xi = \sum_{j=1}^m \xi_j \otimes e_j$, use the finite space

$$E = \operatorname{span}\{A^k \xi_j : 0 \le k \le d,\ 1 \le j \le m\}.$$

For $A_E = P_E A|_E$, induction gives $A_E^k \xi_j = A^k \xi_j$ for $k \le d$. Therefore $v[A_E]\xi = v[A]\xi$, while $W(A_E) \subset W(A)$. Taking the supremum over ξ proves the polynomial bound without a separability assumption.

Finally $K = \overline{W(A)}$ is compact convex and contains the spectrum. To check the last assertion, for $\lambda \notin K$ the numerical range estimate bounds both $\lambda I - A$ and its adjoint below by $\operatorname{dist}(\lambda, K)$. The former has closed range, the latter has zero kernel, so that range is also dense. For a matrix function holomorphic near K, choose K_ϵ inside its domain. Polynomial approximation entrywise on this compact convex set follows from Runge's theorem. On a fixed contour inside K_ϵ enclosing K, the resolvent formula gives convergence in operator norm; uniform convergence on K passes the upper bound to the limit. Approximation only on K, without control on such a contour, would not justify this step.

4 Local metric and optimality certificates

The following results are local additions. They apply to a supplied finite matrix and certified numerical enclosures; they do not prove the source's universal $p(f(A)) \le 4$.

4.1 A repair that includes error in the operator

Theorem 4.1 (Model-aware Stein repair)**.** *Let $\widehat{T}$ approximate T, with $\left\|T - \widehat{T}\right\| \le \delta$. Suppose certified Hermitian matrices $\widehat{H}, K$ and real numbers satisfy*

$$0 < hI \preceq \widehat{H} \preceq uI, \quad 0 \prec K \preceq kI, \quad \mathcal{L}_{\widehat{T}}(\widehat{H}) \succeq -\varepsilon I, \quad \mathcal{L}_{\widehat{T}}(K) \succeq c_0 I,$$

where $\delta, \varepsilon \ge 0$. Set

$$w = 2\left\|\widehat{T}\right\|\delta + \delta^2, \quad e = \varepsilon + uw, \quad c = c_0 - kw.$$

If $c > 0$, then T is stable and

$$H_+ = \widehat{H} + (e/c)K, \qquad H = H_+/h, \qquad U = \frac{u + ek/c}{h} \tag{13}$$

give a feasible point $I \preceq H \preceq UI$, $\mathcal{L}_T(H) \succeq 0$. Consequently $p(T) \le U$. The repaired similarity yields one positive disk density for all coefficient sizes, with condition number at most $\sqrt{U}$.

Proof. For every Hermitian M, expansion with $\Delta = T - \widehat{T}$ gives

$$T^*MT - \widehat{T}^*M\widehat{T} = \widehat{T}^*M\Delta + \Delta^*M\widehat{T} + \Delta^*M\Delta.$$

Its norm is at most $w\|M\|$. Thus $\mathcal{L}_T(\widehat{H}) \succeq -eI$ and $\mathcal{L}_T(K) \succeq cI$. Testing the latter on an eigenvector of T, with eigenvalue λ, gives $(1 - |\lambda|^2)v^*Kv \ge c\|v\|^2 > 0$, hence $\rho(T) < 1$. Linearity proves $\mathcal{L}_T(H_+) \succeq 0$. Since the correction is PSD, the lower bound is hI; the upper bound is $(u + ek/c)I$. Division proves feasibility. Proposition 2.2 applies to $H^{1/2}TH^{-1/2}$. □

For exact T, the Stein series K from Proposition 2.1 gives $c = 1$. If T is Gaussian-rational (real and imaginary parts rational), the linear system $\mathcal{L}_T(K) = I$ has a unique Gaussian-rational solution: the inverse exists because every eigenvalue of the operator $M \mapsto T^*MT$ has modulus less than one. The convergent series proves that solution positive definite. This observation concerns exact arithmetic, not the conditioning or cost of solving the system. A sharper certified lower eigenvalue of H_+ can replace h after repair.

Remark 4.2 (Rank and face boundary). The added term $(e/c)K$ preserves positivity of the metric but does not preserve the rank of its Stein slack. That is deliberate. One must not report the repaired point as belonging to an optimizer's singular slack face. The separate face repair below is the appropriate procedure when a fixed face is required.

4.2 A trace-sensitive dual lower bound

Theorem 4.3 (Dual residual bracket). *Let T be stable. Suppose $X, Y, Z \succeq 0$, $\operatorname{tr} X = 1$, and define the Hermitian residual*

$$E = X - Y - Z + TZT^* = E_+ - E_-, \tag{14}$$

where $E_\pm \succeq 0$ are its positive and negative spectral parts. Then

$$p(T) \;\geq\; L := \max\left\{1, \frac{\operatorname{tr} Y + \operatorname{tr} E_+}{1 + \operatorname{tr} E_-}\right\}. \tag{15}$$

Any certified feasible upper bound U gives $\sqrt{L} \leq \kappa(T) \leq \sqrt{U}$. For exact residual $E = 0$, the objective lower bound is $\operatorname{tr} Y$.

Proof. For any feasible pair (τ, H), expanding the three slack traces gives the exact signed identity

$$\begin{aligned} \tau - \operatorname{tr} Y - \operatorname{tr}(EH) = {} & \operatorname{tr} Y(H - I) + \operatorname{tr} X(\tau I - H) \\ & + \operatorname{tr} Z(H - T^*HT) \;\geq 0. \end{aligned} \tag{16}$$

Since $I \preceq H \preceq \tau I$, $\operatorname{tr}(EH) \geq \operatorname{tr} E_+ - \tau \operatorname{tr} E_-$. Hence $\tau(1 + \operatorname{tr} E_-) \geq \operatorname{tr} Y + \operatorname{tr} E_+$. Take the infimum over feasible pairs and also use $p(T) \geq 1$. □

If an independent bound $p(T) \leq U$ is already known, application to an optimizer also gives

$$p(T) \geq \operatorname{tr} Y + \operatorname{tr} E_+ - U \operatorname{tr} E_- = \operatorname{tr} Y + \operatorname{tr} E - (U - 1) \operatorname{tr} E_-.$$

These formulas fix the residual sign explicitly. A looser alternative is $p(T) \geq \operatorname{tr} Y - U \|E\|_*$, where $\|E\|_*$ is the nuclear norm. Certified enclosures for the spectral traces may be substituted with their correct monotonic directions: a lower bound for $\operatorname{tr} E_+$ and an upper bound for $\operatorname{tr} E_-$ make (15) conservative. Its numerator is nonnegative. None of these bounds follows from an unverified solver status.

Proposition 4.4 (Approximate endpoint support). *For a feasible (τ, H) and the candidates above, set*

$$g = \tau - \operatorname{tr} Y - \operatorname{tr}(EH) \geq 0.$$

Choose any of the three slack–multiplier pairs

$$(A, W) = (\tau I - H, X), \quad (H - I, Y), \quad (\mathcal{L}_T(H), Z).$$

Suppose P is an orthogonal projection and $A \succeq \gamma(I - P)$, with $\gamma > 0$. Then, with $t = g/\gamma$,

$$\operatorname{tr}((I - P)W) \leq t, \qquad \|W - PWP\|_F \leq \sqrt{t^2 + 2\|W\|\, t}. \tag{17}$$

One may replace g by $\bar{g} = \tau - \operatorname{tr} Y - \operatorname{tr} E_+ + \tau \operatorname{tr} E_- \geq g$.

Proof. Each of the three nonnegative traces in (16) is at most g. The slack lower bound proves the trace estimate. In the decomposition $P \oplus (I - P)$, write $W = \left(\begin{smallmatrix} W_{11} & B \\ B^* & C \end{smallmatrix}\right)$. Then $\|C\|_F \le \operatorname{tr} C \le t$. Since $W^2 \preceq \|W\|\, W$, its lower diagonal block gives $\|B\|_F^2 \le \|W\| \operatorname{tr} C$. Now $\|W - PWP\|_F^2 = 2\|B\|_F^2 + \|C\|_F^2$. The bound for $\bar{g}$ is the same trace estimate as above. □

With P an endpoint spectral subspace, this controls leakage away from that subspace. For a zero gap it recovers the source's exact support. A small positive gap does not prove an exact rank or an exact face. The square-root scale is unavoidable in general: for

$$W_s = \begin{pmatrix} s^2 & s \\ s & 1 \end{pmatrix} \succeq 0, \qquad P = \operatorname{diag}(0, 1),$$

the discarded diagonal trace is s^2, but $\|W_s - PW_sP\|_F = \sqrt{s^4 + 2s^2}$.

4.3 Physical coordinates and certified faces

Suppose a full-column-rank synthesis map J represents a retained space, with physical Gram matrix $G_0 = J^*J \succ 0$, and B is the coefficient operator on that space. The coordinate-correct version of (1) is

$$G_0 \preceq H_c \preceq \tau G_0, \qquad B^* H_c B \preceq H_c. \tag{18}$$

Indeed, whitening gives

$$T = G_0^{1/2} B G_0^{-1/2}, \qquad H = G_0^{-1/2} H_c G_0^{-1/2},$$

and transforms these inequalities exactly into (1). Thus every error norm and lower bound in Theorem 4.1 must be evaluated after this whitening, or in the equivalent G_0-weighted norm. Under a change of coordinates $c = C\tilde{c}$, G_0, H_c transform by $C^*(\cdot)C$, and B by $C^{-1}BC$. The optimal value is invariant. Replacing G_0 by the identity in unnormalized coordinates generally changes the question. For example, a diagonally rescaled representation of $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ can have arbitrarily large Euclidean norm while its physical norm remains one.

If J has a kernel, first pass to a specified independent retained basis or an exact quotient on which $G_0 \succ 0$. Adding a small multiple of the identity to a singular G_0 changes the physical model and is not such a quotient. Furthermore a metric certificate for a compression $P_E A|_E$ says nothing about the omitted component $(I - P_E)A|_E$. No full-space assertion follows without controlling that component or proving invariance of E.

Proposition 4.5 (Face-preserving affine repair)**.** *Fix the operator, physical metric, and objective value τ. Suppose the three slack matrices in* (18) *are required to have supports in supplied subspaces with injective basis matrices U_1, U_2, U_3. Write the slacks as $U_i Z_i U_i^*$. In the real vector space $\mathcal{V} = \bigoplus_i \operatorname{Herm}(r_i)$, let $\mathcal{B}Z = b$ be their affine consistency equations, with redundant equations removed consistently so $\mathcal{B}$ is onto. Choose any right inverse $\mathcal{R}$ of $\mathcal{B}$. If $\widehat{Z}_i \succeq \alpha I$ and*

$$\Delta Z = -\mathcal{R}(\mathcal{B}\widehat{Z} - b), \qquad \max_i \|\Delta Z_i\| \le \alpha,$$

then $Z = \widehat{Z} + \Delta Z$ defines an exactly feasible metric whose slacks stay in the prescribed supports. If the last inequality is strict and $\alpha > 0$, each nonempty slack has rank r_i.

Proof. The right-inverse identity gives $\mathcal{B}Z = b$ exactly. Each $Z_i \succeq (\alpha - \|\Delta Z_i\|)I \succeq 0$; congruence gives PSD slacks. The consistency equations reconstruct $H_c = G_0 + U_1 Z_1 U_1^*$ and enforce the other two slacks. Injectivity of U_i preserves the rank of positive definite Z_i. For a zero-dimensional support its block is simply absent. □

This proposition assumes the face is supplied and justified. A floating-point near-kernel is not a facial certificate. If the data and bases are Gaussian-rational, one can use $\mathcal{R} = \mathcal{B}^*(\mathcal{B}\mathcal{B}^*)^{-1}$ in rational real coordinates. On a chosen positive definite coordinate metric $\mathcal{G}$, the corresponding right inverse is $\mathcal{G}^{-1}\mathcal{B}^*(\mathcal{B}\mathcal{G}^{-1}\mathcal{B}^*)^{-1}$. Direct multiplication verifies either formula. This is the relevant specialization of the exact facial repair in [7]; it is not a new general SDP algorithm.

Proposition 4.6 (Rational certificates above the optimum)**.** *For stable Gaussian-rational T, every real $\tau > p(T)$ admits a Gaussian-rational Hermitian H with all three slacks in* (1) *positive definite. If τ is rational, the entire certificate can be rational. This does not assert a rational optimizer or a uniform bit bound.*

Proof. Combine an optimizer (p, H_*) with a strictly feasible pair (τ_s, H_s). A sufficiently small positive convex weight on the latter gives objective less than τ and strict lower and Stein slacks; the unused objective margin makes the upper slack strict too. Let $\mu > 0$ be the minimum of their smallest eigenvalues. Approximate this Hermitian matrix by a Gaussian-rational Hermitian one with error $\eta < \mu/(1 + \|T\|^2)$. The lower and upper slacks lose at most η, and the Stein slack loses at most $(1 + \|T\|^2)\eta$, so all remain positive definite. Rational approximation is dense in this finite real coordinate space. □

The private irrational singleton and weak-infeasibility examples in [7] concern general SDP faces. They warn against demanding exact rational optimal witnesses or unbounded approximate Farkas tests. They do not contradict Proposition 4.6, whose strictly positive objective margin is essential.

5 Positive finite representations with a full bias bound

Truncation of a positive density's Fourier series need not remain positive. Positivity matters here because it certifies all coefficient sizes at once. The following construction preserves it without discarding the approximation error.

Theorem 5.1 (Positive Cesaro boundary certificate)**.** *Let D be a stable contraction. For $N \geq 0$, define*

$$\Phi_{D,N}(\zeta) = I + \sum_{k=1}^{N} \left(1 - \frac{k}{N+1}\right) \left(\zeta^{-k} D^k + \zeta^k (D^*)^k\right). \tag{19}$$

It is a positive trigonometric polynomial with integral I. The same density gives, for every finite m, a unital completely positive, completely contractive map on continuous boundary data,

$$Q_N(h) = \int_{\mathbb{T}} \Phi_{D,N}(\zeta) \otimes h(\zeta)\, d\sigma(\zeta).$$

For analytic $h(z) = \sum_{k\geq 0} z^k C_k$ near $\overline{\mathbb{D}}$,

$$h[D] - Q_N(h) = \sum_{k\geq 1} \min\left\{\frac{k}{N+1}, 1\right\} D^k \otimes C_k. \tag{20}$$

If h is holomorphic near $\{|z| \le r\}$, $r > 1$, and $M_r = \max_{|z|=r} \|h(z)\|$, put $q = r^{-1}$. Then

$$\|h[D] - Q_N(h)\| \le M_r \frac{q(1-q^{N+1})}{(N+1)(1-q)^2}. \tag{21}$$

More generally, a certified estimate $\|D^k\| \le C\theta^k$ for all $k \ge 1$, with $0 < \theta < 1$, replaces the right side by CM_r times the same scalar expression with $q = \theta/r$.

Proof. The scalar kernel

$$F_N(\xi) = \frac{1}{N+1}\left|\sum_{j=0}^{N} \xi^j\right|^2 = \sum_{|k|\le N}\left(1 - \frac{|k|}{N+1}\right)\xi^k$$

is nonnegative with integral one. Convolving Φ_D with F_N proves positivity and the formula (19); integration preserves mass. At every matrix size, a pointwise positive input gives a positive tensor integrand. The isometry proof of Proposition 2.2, with $\Phi_{D,N}$ in place of Φ_D, proves complete contractivity. Fourier integration gives moments

$$\int \zeta^k \Phi_{D,N}\, d\sigma = \begin{cases} I, & k = 0, \\ (1 - k/(N+1))D^k, & 1 \le k \le N, \\ 0, & k > N. \end{cases}$$

Subtract these from the exact moments. Absolute convergence justifies the resulting infinite series (20).

The matrix Cauchy coefficient bound is $\|C_k\| \le M_r r^{-k}$, without any factor depending on m. Using $\|D^k\| \le 1$ reduces the error to

$$M_r \sum_{k\ge 1} \min\{k/(N+1), 1\} q^k.$$

To evaluate the sum, write $\min\{k, N+1\} = \sum_{\ell=1}^{N+1} \mathbf{1}_{\{k \ge \ell\}}$ and sum the geometric series in the order ℓ, k. The result is (21). The certified power bound gives the final variant by the same argument. □

The positive exterior approximation is $\Lambda_N(t) = j(t)\Phi_{D,N}(\psi(t))$, again of mass I. For $h = v \circ f^{-1}$, define

$$\mathcal{Q}_N(v) = \int \Lambda_N(t) \otimes v(G(t))\, d\sigma(t).$$

The theorem bounds $v[A'] - \mathcal{Q}_N(v)$. Returning to A multiplies the error by at most $\operatorname{cond} S$. The existence of some $r > 1$ follows from the analytic collars and the neighborhood of analyticity of v; a quantitative estimate requires an actual certified r and M_r. Neither $\mathcal{Q}_N$ nor Q_N is in general multiplicative. They are positive approximations to a holomorphic calculus, not exact homomorphisms.

Corollary 5.2 (Positive nodes and an explicit quadrature remainder)**.** *Choose an integer $L > N$, set $\zeta_\ell = e^{2\pi i \ell/L}$, and*

$$W_\ell = \Phi_{D,N}(\zeta_\ell)/L, \qquad Q_{N,L}(h) = \sum_{\ell=0}^{L-1} W_\ell \otimes h(\zeta_\ell).$$

Then $W_\ell \succeq 0$, $\sum_\ell W_\ell = I$, for all coefficient sizes simultaneously. If h is a polynomial of degree $d < L - N$, then $Q_{N,L}(h) = Q_N(h)$. Under the analytic hypotheses of Theorem 5.1,

$$\|h[D] - Q_{N,L}(h)\| \le M_r \frac{q(1-q^{N+1})}{(N+1)(1-q)^2} + \frac{2M_r q^{L-N}}{1-q}, \qquad q = r^{-1}. \tag{22}$$

For the exterior problem the corresponding nodes are $t_\ell = \psi^{-1}(\zeta_\ell)$, with the same weights W_ℓ.

Proof. Root-of-unity averaging integrates every Laurent monomial whose nonzero exponent has modulus less than L. This proves the mass identity, since $\Phi_{D,N}$ has degree $N < L$. The product with h has exponents between $-N$ and $N + d$; the asserted degree condition gives exact integration. Both maps are complete contractions because their positive weights sum to I. Truncate the analytic series at degree $d = L - N - 1$. Its uniform error on $\mathbb{T}$ is at most $M_r q^{L-N}/(1-q)$. The two contractive maps differ by at most twice this error and agree on the truncation. Add (21). The exterior node formula is the change of variables already used in (5); no extra factor j is put into these discrete weights. □

This is a finite positive certificate for bounded analytic data with an explicit error. It is not a claim that finitely many evaluations reproduce all analytic functions. Without a collar bound on the data, the displayed quantitative error is unavailable.

6 Exact examples and failure tests

6.1 Optimality with singular Stein slack

Let

$$T_a = \begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}, \quad a > 1, \quad H_* = \operatorname{diag}(1, a^2), \quad X = \operatorname{diag}(0,1),\ Y = \operatorname{diag}(a^2, 0),\ Z = \operatorname{diag}(0,1).$$

Then $\mathcal{L}_{T_a}(H_*) = \operatorname{diag}(1,0)$, $X - Y = Z - T_a Z T_a^*$, $\operatorname{tr} X = 1$, and $\operatorname{tr} Y = a^2$. Thus primal and dual certificates agree:

$$p(T_a) = a^2, \quad \kappa(T_a) = a, \quad D = H_*^{1/2} T_a H_*^{-1/2} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Alternatively, the lower bound follows directly by testing the Stein inequality on e_2: $H_{22} \ge a^2 H_{11} \ge a^2$. When $1 < a < 2$, the numerical range is the disk of radius $a/2 < 1$. Taking $\Omega = \mathbb{D}$, $f(z) = z$, this is an admissible source example with strict $\kappa < 2$. As $a \uparrow 2$, it approaches the sharp endpoint. No universal improvement results.

Despite $\rho(D) = 0$, $\|D\| = 1$. Formula (3) becomes

$$\Phi_D(\zeta) = \begin{pmatrix} 1 & \zeta^{-1} \\ \zeta & 1 \end{pmatrix},$$

of rank one at every point and mass I. For $N \ge 1$, its Cesaro approximation has off-diagonal coefficient $N/(N+1)$ and is positive definite. For $h(z) = zI_m$, the error norm is exactly $1/(N+1)$. Thus positivity and mass do not imply exact moments, and even a nilpotent matrix can have a nonzero Cesaro bias. Positive smoothing also need not preserve the original density rank.

6.2 An exact repair and a deliberately incorrect truncation

Take $0 < \varepsilon < a^2 - 1$ and $\widehat{H} = \operatorname{diag}(1, a^2 - \varepsilon)$. Its Stein slack is $\operatorname{diag}(1, -\varepsilon)$. The exact Stein solution is $K = I + T_a^* T_a = \operatorname{diag}(1, 1 + a^2)$. The repair $\widehat{H} + \varepsilon K$ equals $(1 + \varepsilon)\operatorname{diag}(1, a^2)$. Renormalizing by its actual lower eigenvalue $1 + \varepsilon$ recovers the optimizer. This is an exact instance, not evidence that arbitrary repairs are optimal.

For the scalar contraction $D = r$, with $1/2 < r < 1$, the raw degree-one Fourier truncation of Φ_D at $\zeta = -1$ equals $1 - 2r < 0$. In contrast, its Cesaro approximation is $1 - r > 0$. This supplies a concrete reason for the weights in (19). Replacing the infinite series by an unweighted finite Fourier sum would invalidate positivity and the dimension-uniform certificate.

6.3 What numerical tests can and cannot certify

The accompanying local script checks exact rational primal/dual identities, the residual sign and spectral-part bound on exact fixtures, physical congruences, face repair, polynomial moments, and the finite geometric sum. Additional seeded numerical cases check the ordered density congruence, rank-one example, positive node sums, and model-error inequalities. Exact arithmetic checks validate the stated examples; floating tests are diagnostic only. The general statements are justified by the proofs above, not by a random sample or a claim about solver success. Test counts, results, script hash, final source hash, and PDF evidence are in the owned audit receipts, so this paragraph need not embed a circular hash of its own final file.

7 Private proof lineage and contribution boundaries

7.1 Actual transfers from the routed originals

The full 460-line private source [7] was read, including `thm:face`, `thm:repair`, `thm:weak`, and `thm:irrational`. Its zero-trace facial support proof supplies the exact distinction between a certified face and an approximate kernel. Its right-inverse repair becomes Proposition 4.5; its square-root leakage example explains the scale in Proposition 4.4. The irrational singleton and weak-infeasibility examples are obstructions, not algorithms for the present SDP. Its supplied-certificate verification bounds are not bounds for finding an optimal certificate.

The original proofs in [8] of `thm:neumann`, `thm:frame`, `prop:schurresidual`, and `thm:woodbury` were read with their local hypotheses. The relevant lesson is that a specified norm, retained space, and full residual are part of a certificate. A spectral radius is not a Neumann contraction factor, a frame statement on a retained subspace is not a full-space statement, and a Schur residual must include its eliminated-coordinate term. No graph theorem or Woodbury update is asserted for this report's SDP merely because both involve matrices. The finite positive quadrature here has its own proof and is not an application of a graph sampling theorem.

The complete 210-line physical-metrics chapter [9] was read. Its Gram normalization and omitted-component residual identity motivate the exact retained-space qualifications in Section 4.3. Metric, covariant matrix, and coefficient operator are distinct objects. The present report preserves this distinction and makes no continuum claim from finite measurements. These private sources are not uploaded or included in public search.

7.2 Advantages, limitations, and next work

The useful gain over the inherited structural statement is an auditable route from a candidate matrix metric to upper and lower condition-number bounds, together with a finite positive boundary approximation whose error survives at every matrix coefficient size. The dual bound is sensitive to positive and negative residual traces rather than only a worst-case operator norm. Face-preserving and interior repairs are explicitly different procedures.

The limitations are substantive. Conformal maps and their certified collars remain inputs for the exterior problem. Model error in $T = f(A)$ includes the error of evaluating f, not just the error in A. A Stein solve may be badly conditioned. Approximate endpoint support does not certify exact rank. Positive finite approximations are not exact functional calculi, and their analytic bias bound requires data on a larger disk. None of these local certificates verifies the entire source globally or establishes priority over classical SDP duality, Stein theory, or Cesaro approximation.

Feasible next steps are interval-certified conformal evaluation, exact or interval LDL certificates for the supplied slacks, and a numerically stable implementation of the positive nodes with both metric and quadrature errors reported. Another bounded question is whether extra structure of $f(A)$ permits smaller certified negative residual trace without obscuring coordinate dependence. Finding or proving an optimal singular face automatically is a separate problem; it is not solved by the conditional face repair.

The assessment is C3/5 because the local results combine actual source structures with proved, usable certificate and approximation bounds. It is not C4 or C5: the universal complete Crouzeix bound, optimal-similarity existence, separation construction, and common exact density are inherited. Applicability level L4 is a routing assessment, not this contribution score.

A Read scope and source identity

The actual public-source read covers `main.tex` lines 1–1523, `references.tex` lines 1–125, `figures/coordinates.tex` lines 1–36, and the complete preprint README. The figure's TeX was read; the source PDF and rendered source figure were not visually audited. The proof focus was: SDP and similarity (277–313), density (316–374), separation and extremal realization (376–534), exterior kernels and projection (536–830), comparison (831–1065), geometric and Hilbert-space limits (1067–1309), and conformal appendix (1311–1523). These are actual line ranges, not a claim of formal verification of every inference.

Private reads were: `TR-item-305-astra.tex` all lines 1–460; `TR-core-pat-5.tex` lines 1–38, 70–119, 189–239, 249–308, and 364–462; and `ch01d-physical-metrics.tex` all lines 1–210. Transitive references and the physical-metrics figure asset were not independently read. The exact file paths, hashes, and line ledger accompany the audit.

Public literature checking included the two versioned arXiv metadata pages, the statement of Paulsen's completely bounded similarity theorem, Stein's two-page paper, and Delyon–Delyon Section 2, including its stable-contraction hypothesis and normalized positive measure. These are limited primary-source checks, not a full audit of those papers or of all source references. No scalar result was silently promoted to a complete one.

The frozen upstream comparison records the M631 directory unchanged between the catalogue pin and its October 8 upstream snapshot: five tracked blobs include the public PDF, whereas the local

substantive text identity below covers four text files. Byte identity is evidence of version identity only, not mathematical validity. The base commit is `adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

SHA-256, public main source

`8a5de532bd3bd4c0b84cc8114c22175968813a6e984ced0d1bf48011270e621c`

SHA-256, references

`f3b98c87d0d9f657061b21b80d14718fb211b366c60ad0d56af3a493f4dbc56b`

SHA-256, coordinate figure TeX

`0eddcfce916eecbe737f97bc52c728bb5693050a2803131056baf5cff2a3cf04`

SHA-256, preprint README

`56858fa983870da1753c3365395039c88dc1a72ff12e70e878a3a542dd8520ee`

SHA-256, TR-item-305-astra

`0df8b3f4c235ff1d4969d757abbd1e25f1de2ea6051e663ed4986bbcfc97fef5`

SHA-256, core pattern 5

`8e7250be4aa3874b2d03386a78cbe6ffc432785469e3af70924e5d93d9bf5183`

SHA-256, physical metrics

`7bc5898243373ee2e45551aa78991808fa020148dc6d07c91c41351b5bb84c5c`

References

[1] OpenAI, *The complete Crouzeix theorem: optimal similarity and a common positive boundary representation*, September 23, 2026. OpenAI Math Release, source directory `https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/The-complete-Crouzeix-theorem-September-23-2026`. Local source identity and read scope are in Appendix A.

[2] E. Lorist and F. L. Schwenninger, *A solution to Crouzeix's conjecture*, arXiv:2608.03841v2, August 17, 2026. Versioned metadata checked October 8, 2026.

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[7] Integral Research, *Exact Rational-Data Semidefinite Feasibility: Facial Certificates, Metric Repair, and Bit-Model Obstructions*, private technical report TR-item-305-astra, September 25, 2026. Actual local original read; not a public novelty certificate.

[8] Muralidhara Subbarao, *Adaptive Local–Global Graph Operators with Certified Updates and Reconstruction*, private technical report TR-core-pat-5, October 1, 2026. Selected original proofs read.

[9] Muralidhara Subbarao, *A Nonorthogonal Representation Carries a Metric*, private Volume 21 chapter, `ch01d-physical-metrics.tex`. Complete chapter source read; no publication date inferred from filesystem metadata.